

Chengye Wang

Master's Student in Mechanical Engineering
Beijing Institute of Technology, China



 Website

 leowang1100@gmail.com

 +86 166 4309 0905

 LinkedIn

 Google Scholar

 GitHub

★ PROFILE SUMMARY

- ◆ I got a Bachelor's degree in Vehicle Engineering as an outstanding graduate at Jilin University, China. I am now pursuing a Master's degree in Mechanical Engineering at Beijing Institute of Technology, China.
- ◆ My research experiences focus on path planning and dynamic control of a four-wheel independent-drive (4WD) wheeled robot. I am currently concentrating on improving the path-tracking capability of vehicles under extreme conditions using a combination of vehicle dynamics and learning-based methods.
- ◆ I have a solid grasp of the fundamental theories of vehicle kinematics and dynamics and basic knowledge of trajectory planning and control systems. I am proficient in Matlab/Simulink, CarSim, Python, C++, and LaTeX. I am also familiar with Ubuntu/ROS, Vehicle Control Units (VCUs), Hardware-in-the-Loop (HiL) platforms, CAN and EtherCAT communication, motor control, navigation, and positioning modules.
- ◆ My future research interest lies in combining model-optimization theory with learning-based technologies to improve the motion performance and decision-making behavior of autonomous vehicles in complex environments.

★ EDUCATION

Beijing Institute of Technology (Project 985), Beijing, China

Sep. 2024 – Jun. 2027

Master of Mechanical Engineering

Supervisor: Yechen Qin

- ◆ Research interests: stability analysis, motion control, trajectory planning, and state estimation.
- ◆ Thesis: the thesis topic is yet to be determined.
- ◆ Main courses: Vehicle Dynamics, Numerical Analysis, Advanced Control, and Modern Control Theory.
- ◆ Awards, and scholarships: Outstanding Freshman Scholarship, 4th in Urban and Highway Tracks of Onsite Autonomous Driving Algorithm Challenge

Jilin University (Project 985), Changchun, China

Sep. 2020 – Jun. 2024

Bachelor of Vehicle Engineering

Supervisor: Nan Xu

- ◆ GPA: 3.73/4.0 (top 5%).
- ◆ Thesis: "Vehicle trajectory method planning based on polynomial curve optimization."
- ◆ Main courses: Calculus, Linear Algebra, Automobile Theory, Principles of Control Engineering.
- ◆ Awards, and scholarships: Outstanding Graduate.

★ PUBLICATION

Chengye Wang, Shaoyang Shi, Yu Zhang, and Yechen Qin. "Controlled Stability Region-Guided Lateral-Longitudinal Trajectory Tracking for Autonomous Vehicles." IEEE Transactions on Industrial Electronics. (**Under review, IF=7.4, JCR Q1, Top**)

Chengye Wang, Yu Zhang, Xuepeng Hu, Shaoyang Shi, Jingqi He, and Yechen Qin. "Coordinated Tracking and Stability Control of Automated Vehicles at the Limits: A Controlled Stability Region Approach." 2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), Naples, Italy. IEEE. (**Accepted**)

Chengye Wang, Yu Zhang, Xuepeng Hu, Haipeng Qin, Guoli Wang, and Yechen Qin. "Real-time multidimensional vehicle dynamic stability domain calculation and its application in intelligent vehicles." ICVS 2025, Chongqing, China. SAE International. (**Accepted**)

Yu Zhang, **Chengye Wang**, Fu Du, Mingming Dong, Yechen Qin, and Ming Mao. "Dynamic Region of Stability Integrated Path-tracking Control for Intelligent Vehicles." Chinese Journal of Mechanical Engineering. (**Accepted**)

Xuepeng Hu, Yu Zhang, **Chengye Wang**, Zhenfeng Wang, and Yechen Qin. “MCTP: A Multi-Coupled Dynamics Trajectory Planning Scheme for Autonomous Driving in Extreme Conditions.” IEEE Transactions on Automation Science and Engineering. (Accepted, IF=6.4, JCR Q2, Top)

Shaoyang Shi, Yu Zhang, Kai Yang, **Chengye Wang**, Zhenfeng Wang, and Yechen Qin. “A Risk-Aware Decision Making and Motion Planning Framework for Interactive Autonomous Driving.” 2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC), Naples, Italy. IEEE. (Accepted)

Jingqi He, Ziyi Yang, Shaoyang Shi, **Chengye Wang**, Yu Zhang, and Yechen Qin. “Tire Model-Free Tire-Road Friction Coefficient Estimation for Unmanned Platforms via Vision-Dynamics Dual-Confidence Fusion.” Mechanical Systems and Signal Processing. (Under review, IF=10.2, JCR Q1, Top)

★ RESEARCH EXPERIENCE

Graduate Student & Vehicle Engineer

May 2025 – Present

Vehicle electrical-system setup, controller algorithm design, and real-vehicle testing

Xinxiang Northern Vehicle Instrument Co., Ltd., China

- ◆ Design the driving, braking, and steering controllers in the VCU using Matlab/Simulink and implement communication between the vehicle controller and actuators using CAN and EtherCAT.
- ◆ Assist in building the complete vehicle electrical system.
- ◆ Conduct vehicle testing at speeds up to 50 km/h and on slopes of up to 30°. Implement manual and remote-control modes and active pose control.

National Natural Science Foundation of China

Sep. 2024 – Present

Vehicle lateral stability analysis considering the combined effects of multiple factors and real-time coordinated control

Beijing Institute of Technology, China

- ◆ Analyze nonlinear lateral stability in the phase plane while considering vehicle static parameters, driving conditions, and active control capability using the Lyapunov function method and phase portrait method.
- ◆ Quantify the stability region under real-time driving conditions and use it to coordinate trajectory tracking and stability control.
- ◆ Design longitudinal-lateral coupled trajectory-tracking and stability controllers based on Model Predictive Control.

National Natural Science Foundation of China

Nov. 2024 – Present

Model-data-driven trajectory planning for interactive conditions and real-vehicle testing

Beijing Institute of Technology; BYD Auto Co., Ltd., China

- ◆ Develop a multi-coupled dynamics trajectory-planning scheme that considers lateral-longitudinal motion coupling, tire-force coupling, and lateral instability for autonomous driving under extreme conditions.
- ◆ Develop a risk-aware decision-making and motion-planning framework in which a Soft Actor-Critic agent generates high-level interaction commands that are embedded into an MPC module for constrained trajectory optimization with safety and dynamic guarantees.
- ◆ Design the Hardware-in-the-Loop platform and conduct cooperative real-vehicle testing with BYD.

★ LANGUAGE & SKILLS

- ◆ **English:** CET-6 (China’s College English Test Band): 544 (passed); writing, reading, and speaking.
- ◆ **Chinese:** Native language.
- ◆ **Research skills:** Matlab/Simulink, CarSim, Python, Ubuntu/ROS, C++, LaTeX, automotive hardware and software debugging, and Microsoft Office.

★ REFEREES

Dr Yu Zhang

Assistant Researcher

China North Vehicle Research Institute, China

Email: zhyss_bit@bit.edu.cn

Professor Yechen Qin

School of Mechanical Engineering

Beijing Institute of Technology, China

Email: qinyechen@bit.edu.cn